Top 50 SQL Interview Questions

Practice checklist • Questions only • No solutions

Basic SQL Queries

1. Write a query to fetch all records from an employee table.
2. Write a query to fetch employees whose salary is greater than 50,000.
3. Write a query to find employees whose names start with the letter 'A'.
4. Write a query to find employees whose salary is between 30,000 and 70,000.
5. Write a query to fetch distinct department IDs from the employee table.
6. Write a query to sort employees by salary in descending order.
7. Write a query to find the total number of employees.
8. Write a query to find the maximum, minimum and average salary.
9. Write a query to find employees hired after a particular date.
10. Write a query to fetch the first 5 highest-paid employees.

GROUP BY and Aggregate Functions

11. Write a query to find the department-wise employee count.
12. Write a query to find the department-wise average salary.
13. Write a query to find the department-wise maximum salary.
14. Write a query to find departments having more than 5 employees.
15. Write a query to find departments where the average salary is greater than 50,000.
16. Write a query to count employees based on their job titles.
17. Write a query to find the total salary expenditure for each department.
18. Write a query to find departments having fewer than 3 employees.

SQL JOIN Questions

19. Write a query to display employee names along with their department names.
20. Write a query to find employees who do not belong to any department.
21. Write a query to find departments that have no employees.
22. Write a query to display all employees and their department details, including employees without departments.
23. Write a query to find employees working in the same department.
24. Write a query to find employees earning more than their department manager.
25. Write a query to display employee names and their manager names using SELF JOIN.
26. Write a query to find employees who work in the same department as a given employee.

Subqueries and Ranking

27. Write a query to find the second-highest salary.
28. Write a query to find the third-highest salary.
29. Write a query to find the Nth-highest salary.
30. Write a query to find the highest-paid employee in each department.

31. Write a query to find the second-highest salary in each department.
32. Write a query to find employees earning more than the overall average salary.
33. Write a query to find employees earning more than their department's average salary.
34. Write a query to find the top 3 highest-paid employees in each department.
35. Write a query to find the department with the highest average salary.

Duplicate and Data Cleaning Queries

36. Write a query to find duplicate employee names.
37. Write a query to find duplicate email addresses.
38. Write a query to delete duplicate records while keeping one record.
39. Write a query to find records with NULL values.
40. Write a query to replace NULL salary values with zero.
41. Write a query to find employees whose salary is duplicated.
42. Write a query to remove leading and trailing spaces from employee names.

Advanced SQL Queries

43. Write a query using ROW_NUMBER() to rank employees by salary.
44. Write a query to demonstrate the difference between RANK() and DENSE_RANK().
45. Write a query to calculate the running total of employee salaries.
46. Write a query to calculate the cumulative average salary.
47. Write a query to find the highest salary in each department using window functions.
48. Write a query to find employees hired in the same year.
49. Write a query to find the month-wise employee joining count.
50. Write a query to find employees whose salary is greater than the previous employee's salary using LAG().